



Hotel Management System

A comprehensive Tkinter-based application for managing hotel operations, from room bookings to restaurant services

What This Application Does



Room Management

Handles room availability, booking, and checkout processes with real-time tracking



Customer Details

Collects and stores essential guest information including name, phone, address, and ID



Payment Processing

Supports multiple payment methods: PhonePe, Card, and Cash transactions



Restaurant Service

Integrated food ordering system with menu items and table booking options

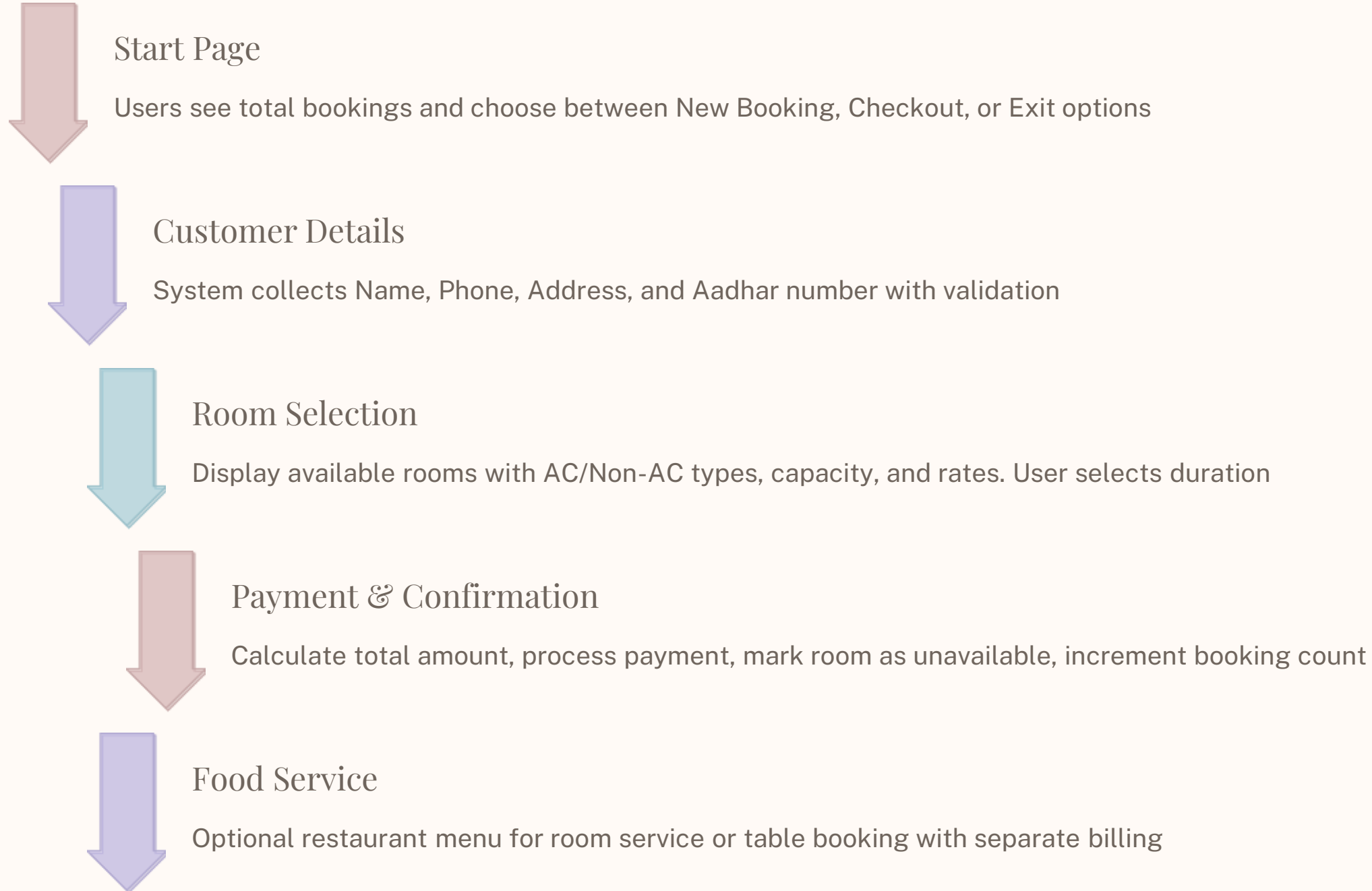
System Architecture Overview

Core Data Structures

The application uses Python dictionaries to manage hotel resources:

- **rooms** dictionary stores 6 rooms with properties like type, capacity, availability, and rates
- **menu_items** dictionary contains 9 food items with pricing
- **current_booking** tracks active reservation details
- **total_bookings** counter maintains booking statistics

Application Flow: Step-by-Step



Room Inventory Details

Room No.	Type	Capacity	Rate/Day	Status
101	AC	2 People	₹1,500	Available
102	AC	4 People	₹2,200	Available
201	Non-AC	2 People	₹1,000	Available
301	AC	3 People	₹2,000	Available

The system dynamically updates room availability status after each booking and restores it during checkout

Key Technical Components

GUI Environment Check

The application begins with a critical validation that checks for graphical display availability on non-Windows systems. If no `DISPLAY` environment variable exists, the program exits gracefully with a clear error message directing users to run it on a proper desktop environment.

Window Management

The `clear_window()` function is essential for navigation, destroying all existing widgets before rendering new screens. This creates a single-window experience where each section smoothly replaces the previous one, maintaining a clean 500x600 pixel interface.

Checkout System

The checkout process validates room occupancy, calculates extra charges for extended stays beyond the booked period, and automatically marks rooms as available again. It uses `simplifiedialog` for quick integer input of room numbers and actual stay duration.

How Output is Generated

Tkinter Rendering Process

The application uses Tkinter's event-driven architecture:

1. **Root Window Creation:** `tk.Tk()` initializes the main window with 500x600 dimensions
2. **Widget Placement:** Labels, Buttons, and Entry fields are packed vertically using `.pack()` method
3. **Event Loop:** `root.mainloop()` keeps the window responsive, waiting for user interactions
4. **Callback Functions:** Button clicks trigger navigation functions that rebuild the interface

Each user action triggers a specific function (like `customer_details()` or `payment_page()`) that clears the current screen and renders new widgets dynamically

Conclusion & Key Takeaways



Simple Yet Functional

This application demonstrates how a basic Tkinter GUI can manage complex hotel operations including bookings, payments, and food service without requiring a database



State Management

Using dictionaries and global variables (`rooms`, `current_booking`, `total_bookings`) provides a straightforward approach to tracking system state across multiple screens



Enhancement Opportunities

While functional, this code could benefit from data persistence (saving bookings to files/database), better error handling for edge cases, and separation of UI and business logic into different modules

Food Service

Order to Room

Book a Table

Skip

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THANK YOU.....

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